

Machine Learning Pipeline for Biomedical Text Analytics Using Orange

"Analyzing Biomedical Text with ML"

Presented by Anjali Saini

M.Sc. Bioinformatics

Maharishi Dayanand University, Rohtak



Automating Biomedical Text Analysis

The rapid increase in biomedical literature necessitates automated analysis methods. This project leverages **Orange Data Mining** to construct a comprehensive text-mining workflow, from preprocessing to classification.

The Challenge

Manual analysis of vast biomedical text is inefficient and prone to human error, requiring advanced computational solutions.

Our Solution

Integrating NLP and ML techniques to automate text classification, enhancing discovery and research efficiency.

Orange Data Mining

A visual, no-code platform that simplifies complex data analysis workflows, making it accessible for diverse users.

Project Objectives

Our primary goals involved a complete machine learning pipeline for analyzing 400 biomedical abstracts.



Text Preprocessing



TF-IDF Conversion



PCA Application



Hierarchical Clustering



ML Model Training & Evaluation



Unseen Data Prediction

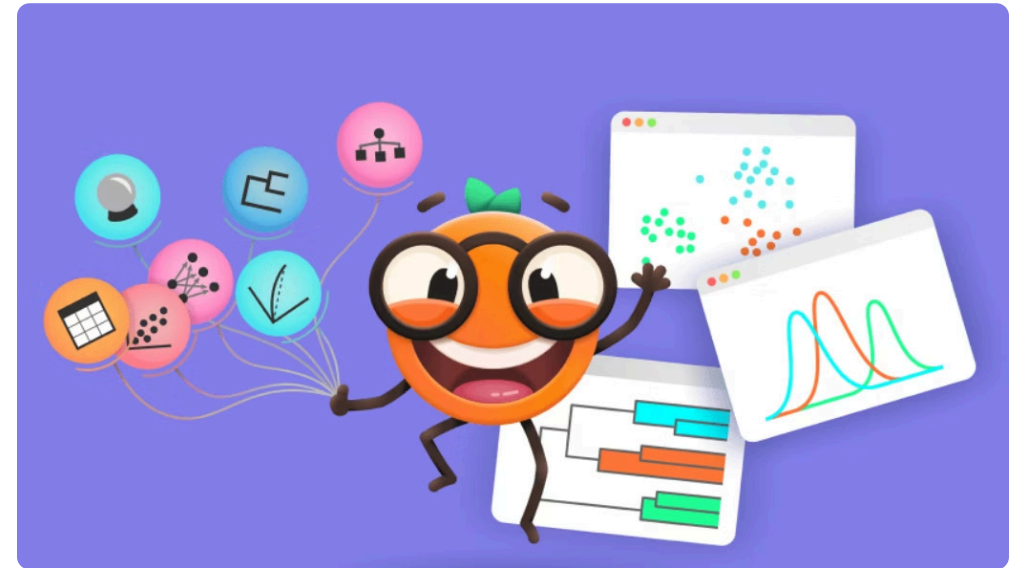
Comparison of Tools for Biomedical Text Mining

Feature / Tool	Orange	Python	KNIME
Ease of Use	★★★★★ (No coding, drag-and-drop)	★★ (Coding required)	★★★★ (Low-code workflows)
Text Mining Support	Strong (Text Mining Add-on)	Very Strong (NLTK, spaCy, scikit-learn)	Strong (Text Processing Extension)
Machine Learning Models	Built-in, easy to use	Full control, all ML models	Wide library, similar to Orange
Visualization	Excellent, instant	Requires coding	Very good
Learning Curve	Very Easy	Medium-Hard	Moderate
Pipeline Speed	Very Fast	Medium	Fast
Reproducibility	High (workflow files)	Depends on user code	High (workflow based)
Best For	Education, quick ML/NLP	Advanced research	Enterprise workflows

Why Orange Data Mining?

Orange stands out as an ideal platform for this project due to its user-friendly interface and powerful capabilities.

- **No Coding Required:** Enables rapid prototyping and deployment without extensive programming knowledge.
- **Drag-and-Drop Workflow:** Simplifies complex analytical tasks with intuitive visual programming.
- **Built-in Text Mining Tools:** Comprehensive suite for NLP, text processing, and feature extraction.
- **Easy Model Comparison:** Facilitates quick evaluation and selection of the best-performing models.
- **Real-time Visualizations:** Offers immediate feedback and insights into data patterns and model behavior.
- **Educational & Research Friendly:** Perfect for teaching and exploring data science concepts.



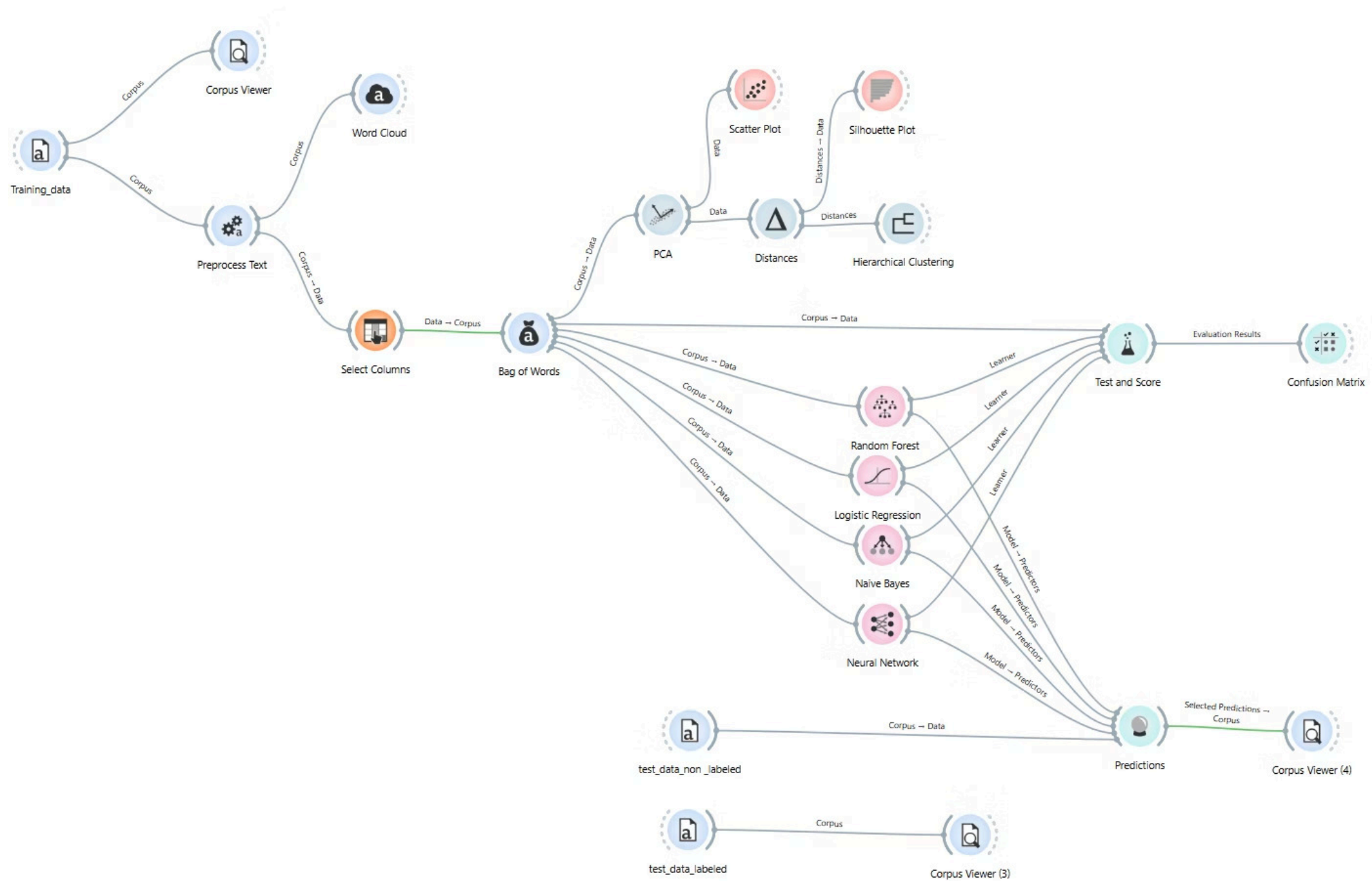
Dataset Overview

Our project utilized a curated dataset of biomedical abstracts, processed through a systematic workflow in Orange.

- **400 Biomedical Abstracts:** Comprehensive collection for robust analysis.
- **8 Distinct Categories:** Covering a variety of biomedical research areas.
- **+10 Unseen Test Abstracts:** For evaluating model generalization on new data.
- **Source:** Primarily from PubMed and Kaggle, ensuring high-quality scientific content.

This diagram illustrates the step-by-step process implemented in Orange, from data ingestion to predictive modeling.

Orange Workflow Diagram



Orange Workflow Components – Part 1



Training_data

Imports labeled biomedical abstracts into Orange.



Preprocess Text

Cleans text using tokenization, stopwords, and lemmatization.



Word Cloud

Visualizes the most frequent words in the corpus.



Select Columns

Keeps only necessary columns (text + label).



Bag of Words (TF-IDF)

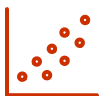
Converts text into numerical TF-IDF feature vectors.



PCA

Reduces dimensionality to reveal structure in text data.

Orange Workflow Components – Part 2



Scatter Plot

Visualizes documents along principal components.



Distances (Cosine)

Calculates similarity between documents.



Test & Score

Evaluates all models using metrics like accuracy and AUC.



Confusion Matrix

Shows category-wise correct vs incorrect predictions.



Predictions

Applies trained models to unseen biomedical abstracts.



Corpus Viewer (Test Data)

Shows predicted labels for test documents.

Preprocessing & TF-IDF Transformation

Preparing raw text for machine learning involves several critical steps to ensure data quality and relevance.



Tokenization

Breaking text into individual words or phrases.



Lowercasing

Standardizing text to uniform case for consistent processing.



Stopword Removal

Eliminating common words that lack significant meaning.



Lemmatization

Reducing words to their base forms to improve accuracy.



Punctuation Removal

Cleaning text by removing non-alphanumeric characters.



Number Filtering

Excluding numerical data unless contextually relevant.

TF-IDF Transformation: After preprocessing, texts were converted into numerical vectors using TF-IDF, generating **7,922 features** for model input.

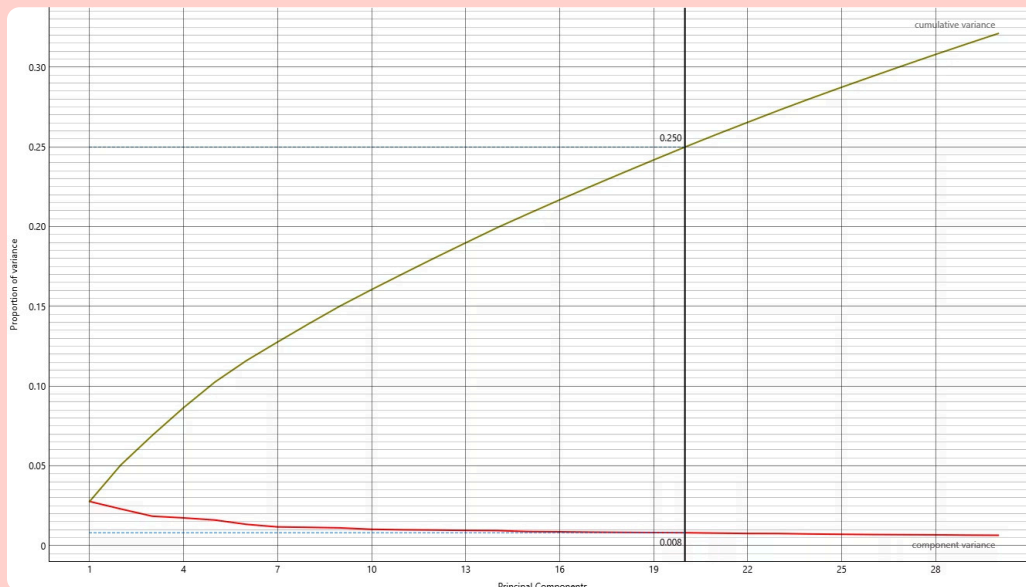
PCA Insights

PCA compressed thousands of text features into meaningful components while preserving essential variance.

PCA Variance Plot

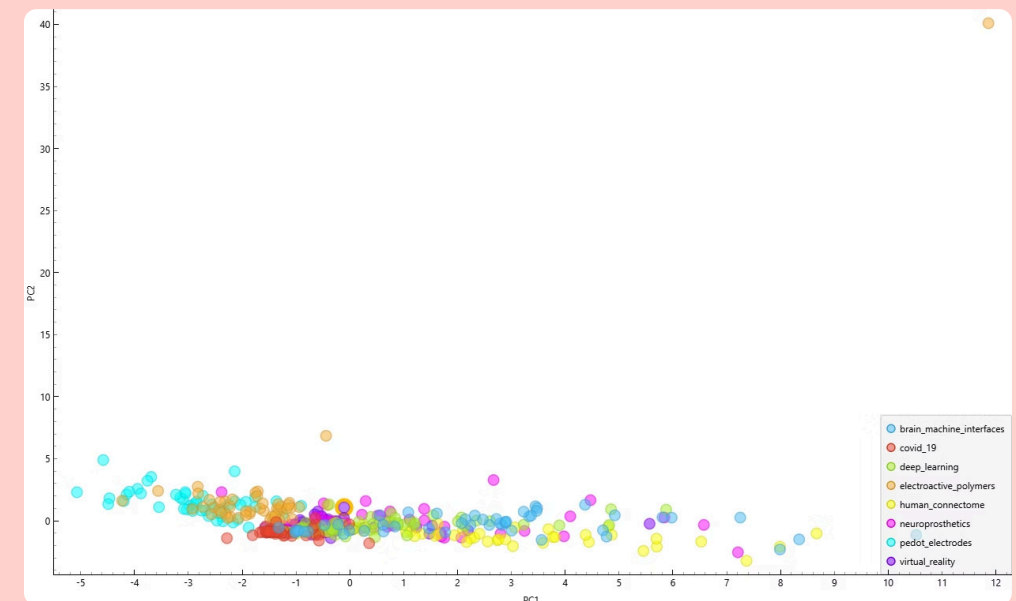
The PCA variance plot shows **how much information (variance)** each principal component captures from the original TF-IDF feature space.

- **PC1:** Explained approximately **2.8%** of the variance.
- **20 PCs:** Accounted for roughly **25%** of the total variance.
- **30 PCs:** Captured around **32%** of the variance, indicating a broad distribution of semantic information.



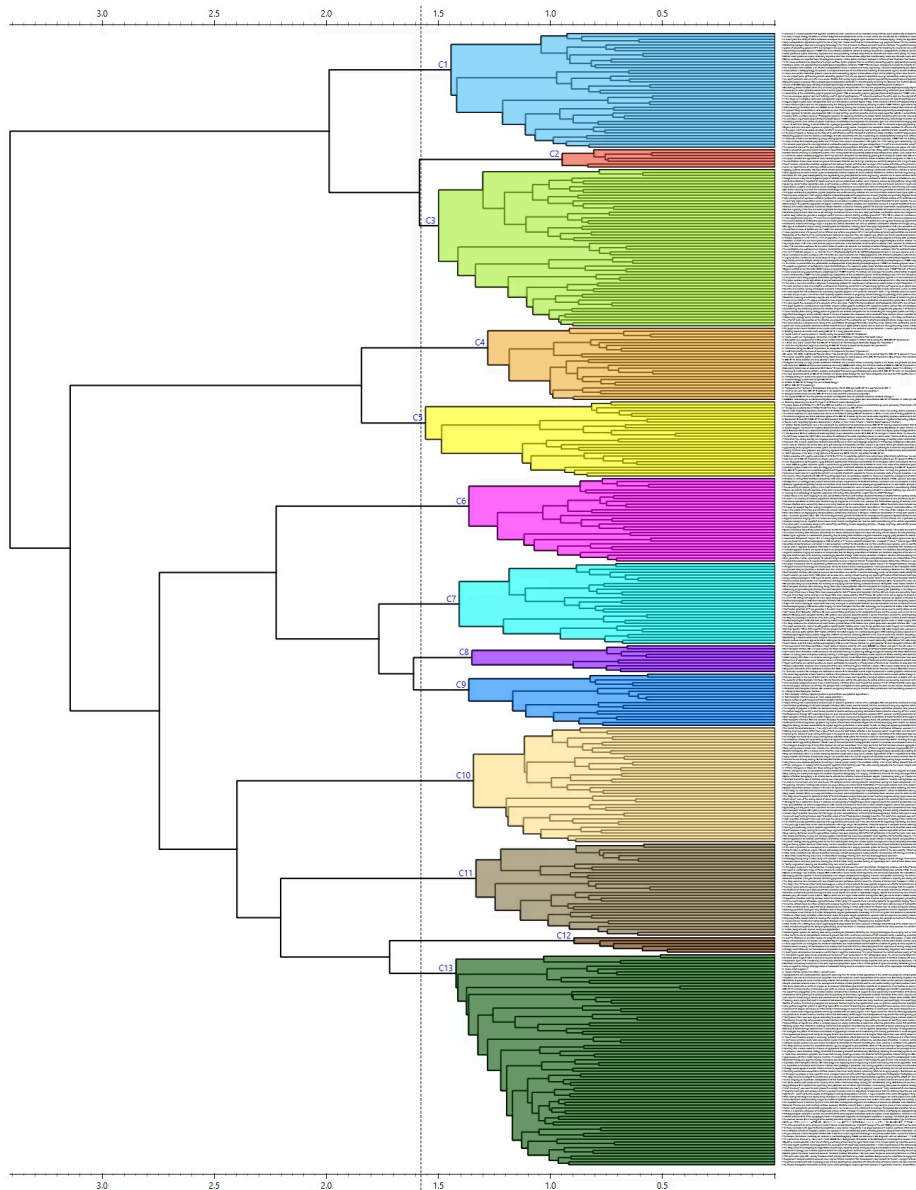
PCA Scatter Plot

The PCA scatter plot visualizes documents along the **first two principal components (PC1 and PC2)**.



- PCA reveals natural grouping of biomedical topics.
- Similar domains cluster together (e.g., COVID, polymers).
- Interdisciplinary fields (deep learning, BMI, connectome) overlap, which is expected.
- Visualization helps confirm cluster structure before applying machine learning models.

Hierarchical Clustering Patterns



Clustering identified natural groupings among the abstracts based on cosine distances:

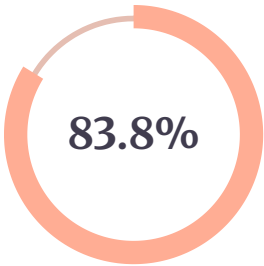
- COVID-19 abstracts form the **most distinct and tightly grouped cluster**.
- PEDOT electrodes and electroactive polymers show **strong material-science clustering**.
- Neural engineering topics (BMI, neuroprosthetics, connectome, deep learning) **overlap because of shared neuroscience vocabulary**.
- Virtual Reality abstracts cluster **close to ML/neuroscience topics**.
- A total of **13 meaningful clusters** identified in the dendrogram.
- Long branches represent **unique or domain-specific abstracts**.

Dendrogram

Illustrating the hierarchical relationships between abstract clusters.

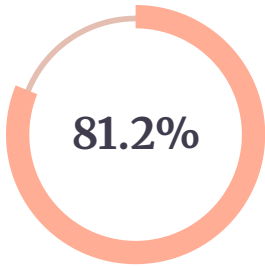
Machine Learning Models & Evaluation

We trained and evaluated four different machine learning models to classify biomedical abstracts, with Logistic Regression emerging as the top performer.



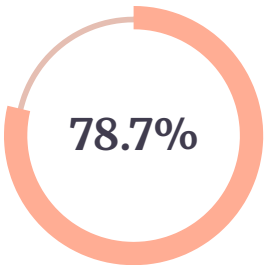
Logistic Regression

Highest accuracy achieved, with an AUC of **0.979**.



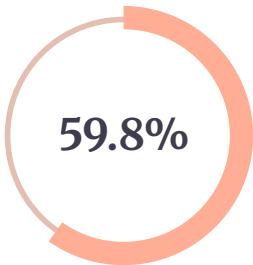
Neural Network

Good at handling nonlinear patterns



Random Forest

Moderate accuracy, doing well on clear categories (like COVID and polymers)



Naive Bayes

A baseline model, Struggles with high-dimensional sparse text data

Test & Score Results

Evaluation results for target (None, show average over classes) ▾									
Model	AUC	CA	F1	Prec	Recall	MCC	Spec	LogLoss	
Logistic Regression	0.979	0.838	0.835	0.844	0.838	0.816	0.977	0.519	
Random Forest	0.954	0.787	0.773	0.790	0.787	0.761	0.970	1.301	
Naive Bayes	0.939	0.598	0.595	0.746	0.598	0.571	0.943	5.726	
Neural Network	0.959	0.812	0.806	0.839	0.812	0.791	0.973	0.788	

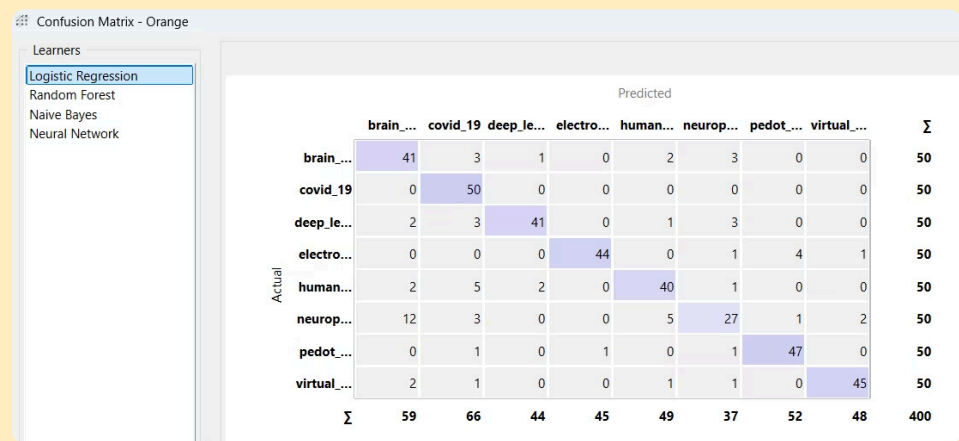
Compare models by: Area under ROC curve ▾ ☐ Negligible diff.: 0.1				
	Logistic Regression	Random Forest	Naive Bayes	Neural Network
Logistic Regression		0.967	0.999	0.988
Random Forest	0.033		0.777	0.334
Naive Bayes	0.001	0.223		0.033
Neural Network	0.012	0.666	0.967	

Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

Confusion Matrices

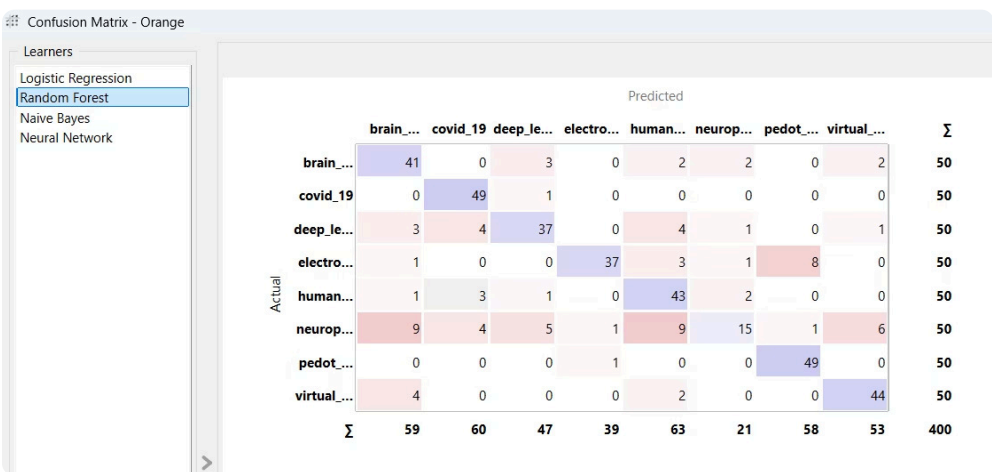
Logistic Regression

- COVID-19, PEDOT electrodes, electroactive polymers, and virtual reality are classified **perfectly or near-perfectly**.
- Small confusion exists between **neuroprosthetics, brain-machine interfaces, and human connectome** due to similar vocabulary.



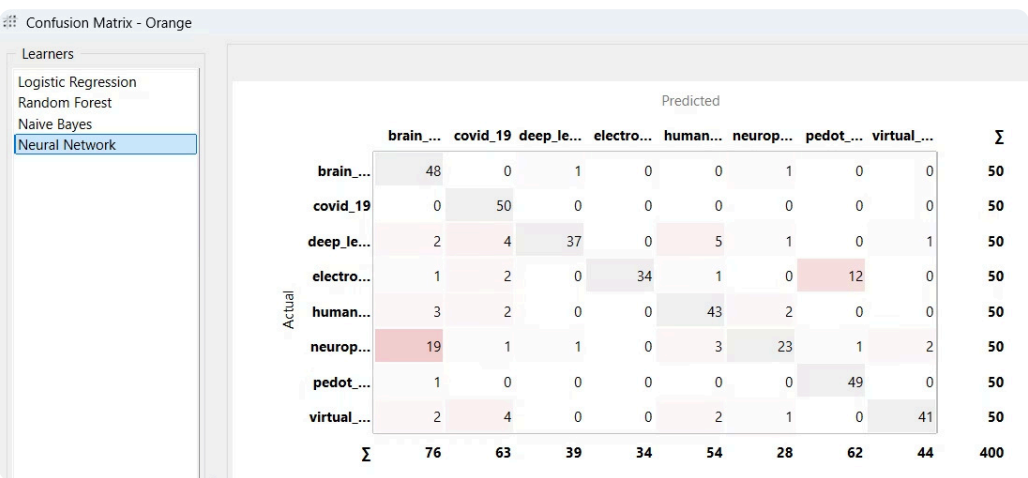
RANDOM FOREST

- strong classification for **material-science categories** (PEDOT, polymers) and **COVID-19**.
- Moderate confusion among **deep learning, BMI, connectome, and neuroprosthetics**.



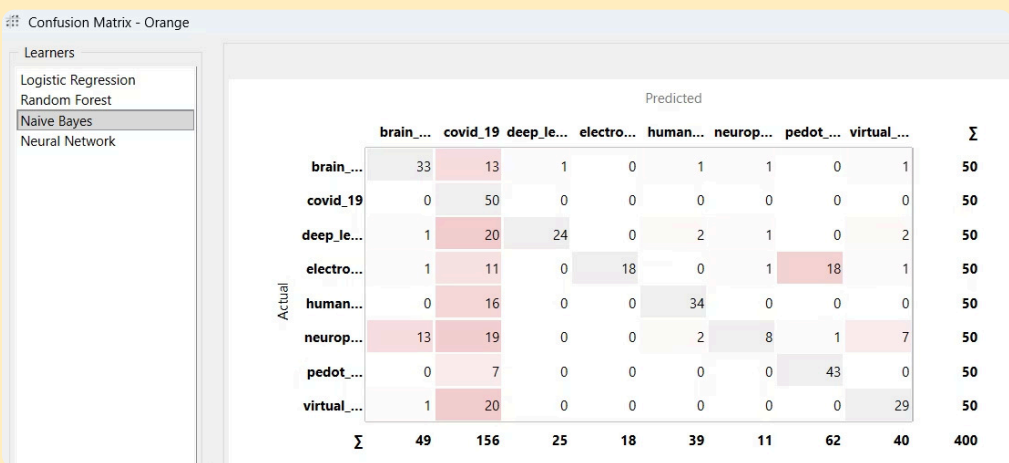
Neural Network

- Performs very well on **connectome, PEDOT, VR, and COVID-19**.
- Shows **slight confusion** between deep learning ↔ neuroprosthetics ↔ BMI.



NAIVE BAYES

- Misclassifies many documents, especially labeling unrelated texts as **"covid_19"**.
- Performs poorly for **neural, VR, and material-science** categories.



Prediction of Unseen Biomedical Abstracts

Predictions - Orange

Show probabilities for (None) v

	Logistic Regression	Random Forest	Naive Bayes	Neural Network	text
1	deep_learning	deep_learning	covid_19	deep_learning	Deep learning approaches have increasingly been e...
2	covid_19	covid_19	covid_19	covid_19	COVID-19 pathogenesis involves a complex interplay...
3	human_connectome	human_connectome	covid_19	human_connectome	Human connectome research aims to map the struct...
4	virtual_reality	virtual_reality	virtual_reality	virtual_reality	Virtual reality technologies are increasingly utilized in...
5	neuroprosthetics	neuroprosthetics	covid_19	neuroprosthetics	Brain-machine interfaces (BMIs) enable direct comm...
6	covid_19	electroactive_polymers	covid_19	electroactive_polymers	Electroactive polymers (EAPs) have emerged as prom...
7	pedot_electrodes	pedot_electrodes	covid_19	brain_machine_interfaces	PEDOT-based electrodes are widely used in neural in...
8	neuroprosthetics	covid_19	covid_19	neuroprosthetics	Neuroprosthetics aims to restore lost sensory or mot...
9	covid_19	covid_19	covid_19	brain_machine_interfaces	Wearable biosensor systems have advanced rapidly ...
10	covid_19	deep_learning	covid_19	deep_learning	Machine-learning-driven biomarker discovery has ac...

- Different model was used to classify 10 unseen biomedical abstracts.
- Majority of predictions matched the expected research domains.
- Strong performance observed for:
 - Deep Learning
 - COVID-19
 - Human Connectome
 - Virtual Reality
 - Neuroprosthetics
- A few mismatches occurred in overlapping categories (e.g., neuroprosthetics vs BMI).
- Results demonstrate strong generalization of the model to new data.

Conclusion & Key Insights



Successfully built a complete machine-learning pipeline in Orange for biomedical text analysis.



TF-IDF and PCA effectively reduced high-dimensional data while preserving semantic structure.



Hierarchical clustering revealed meaningful topic grouping across biomedical domains.



Logistic Regression emerged as the best-performing model with **83.8% accuracy** and **AUC 0.979**.



The workflow generalized well to unseen biomedical abstracts, demonstrating strong prediction capability.



Results confirm the usefulness of no-code platforms like Orange for bioinformatics and NLP research.

Future Scope



Integrate advanced NLP models like **BERT, BioBERT, or PubMedBERT** for deeper biomedical understanding.



Use word embeddings (**Word2Vec, GloVe, FastText**) to capture contextual meaning.



Expand the dataset by collecting thousands of abstracts from **PubMed** for higher model accuracy.



Implement **Named Entity Recognition (NER)** to extract genes, diseases, proteins, and biomarkers.



Apply **topic modeling (LDA or BERTopic)** for discovering hidden research themes.



Deploy the classification model as a **web or desktop application** for real-time biomedical text analysis.

References

This project drew upon a variety of essential resources and literature in the fields of biomedical text mining and machine learning.



- **PubMed:** Primary source for biomedical abstract data.
- **Kaggle PubMed Dataset:** Curated dataset used for initial abstract collection.
- **Orange Documentation:** Official guides and tutorials for Orange Data Mining software.
- **PCA Literature:** Academic papers on Principal Component Analysis for dimensionality reduction.
- **Clustering Literature:** Research on hierarchical clustering and other unsupervised learning techniques.
- **Machine Learning Literature:** Textbooks and journals on Logistic Regression, Neural Networks, Random Forests, and Naive Bayes.



Thank You

We appreciate your engagement and interest in our work on biomedical text mining.